

SAEED ABDULLAH

SAEED@PSU.EDU • SAEEDABDULLAH.COM

- 2024 – **Associate Professor**
College of Information Sciences and Technology
Penn State University
- 2017 – 2024 **Assistant Professor**
College of Information Sciences and Technology
Penn State University

EDUCATION

- 2011 – 2017 **Doctor of Philosophy, Information Science, Cornell University**
- 2009 – 2011 **Master of Science, Computer Science, University of Vermont**
- 2003 – 2007 **Bachelor of Science, Computer Science & Engineering,
Bangladesh University of Engineering & Technology**

SELECTED AWARDS & HONORS

- 2025 **NIH grant for acute stroke detection (Co-I)**
Best Poster Award at MASC-SLL 2025
- 2024 **NIH P30 grant to design health interventions (Co-I)**
NIA R21 grant to support dementia care (PI)
- 2023 **NSF grant to improve mental health worker training (Co-PI)**
NIMH R21 grant on financial behaviors in bipolar disorder (PI)
IST Dean's Circle of Teaching Excellence
- 2020 **Honorable Mention at CHI (top 5%)**
- 2019 **NSF grant on conversational agents for self-management in PTSD (PI)**
IST Junior Faculty Excellence Award in Research
- 2018 **Tronzo Medical Informatics Endowment**
- 2016 **Best Paper Award at MobileHCI (top 1%)**

2013 **\$100,000 Heritage Open mHealth Challenge winner**

PUBLICATIONS

JOURNAL ARTICLES

- 2025 J27 **Artificial Intelligence as a Feedback Teammate for Treatment Delivery: Cognitive Behavioral Therapists' Hopes and Fears**
Lauren Moran, Sadie Kee, Christopher Wiese, Rosa Arriaga, SAEED ABDULLAH, and Andrew Sherrill
Cognitive and Behavioral Practice
DOI: [10.1016/j.cbpra.2025.06.007](https://doi.org/10.1016/j.cbpra.2025.06.007)
- J26 **Effects of Physical Activity on Health Among Informal Caregivers of People With Cognitive Decline: A Systematic Review and Meta-Analysis**
Sangwoo Ahn, Sandra Cobb, Rochelle Butler, SAEED ABDULLAH, Marie Boltz, Misook Chung, Patricia Roberson, Jung-Ah Lee, Xiaopeng Zhao, and Joel Anderson
Western Journal of Nursing Research
DOI: [10.1177/01939459251341197](https://doi.org/10.1177/01939459251341197)
- J25 **There's No "I" in TEAMMAIT: Impacts of Domain and Expertise on Trust in AI Teammates for Mental Health Work**
Nathaniel Swinger, Cynthia Baseman, Myeonghan Ryu, SAEED ABDULLAH, Christopher Wiese, Andrew Sherrill, and Rosa Arriaga
Proceedings of the ACM on Human-Computer Interaction (PACM HCI)
DOI: [10.1145/3710917](https://doi.org/10.1145/3710917)
- J24 **Teaming with Artificial Intelligence to Learn and Sustain Psychotherapy Delivery Skills: Workplace, Ethical, and Research Implications**
Andrew Sherrill, Christopher Wiese, SAEED ABDULLAH, and Rosa Arriaga
Journal of Technology in Behavioral Science
DOI: [10.1007/s41347-025-00484-4](https://doi.org/10.1007/s41347-025-00484-4)
- 2024 J23 **Exploring user engagement with real-time verbal feedback from an exoskeleton-based virtual exercise coach**
Raju Maharjan, Sanjana Mendu, Milton Mariani, SAEED ABDULLAH, and John Hansen
DIGITAL HEALTH
DOI: [10.1177/20552076241302652](https://doi.org/10.1177/20552076241302652)
- J22 **Assessing acceptance and feasibility of a conversational agent to support individuals living with post-traumatic stress disorder**

Hee Jeong Han, Sanjana Mendu, Beth Jaworski, Jason Owen, and SAEED ABDULLAH
DIGITAL HEALTH
DOI: [10.1177/20552076241286133](https://doi.org/10.1177/20552076241286133)

J21 **Using a Smartwatch App to Understand Young Adult Substance Use: Mixed Methods Feasibility Study**
Sahiti Kunchay, Ashley Linden-Carmichael, and SAEED ABDULLAH
JMIR Human Factors
DOI: [10.2196/50795](https://doi.org/10.2196/50795)

J20 **Conversational Agents in Palliative Care: Potential Benefits, Risks, and Next Steps**
Yael Schenker, SAEED ABDULLAH, Robert Arnold, and Kathryn Schmitz
Journal of Palliative Medicine
DOI: [10.1089/jpm.2023.0534](https://doi.org/10.1089/jpm.2023.0534)

2023 J19 **Designing voice interfaces to support mindfulness-based pain management**
Sanjana Mendu, Sebrina Doyle Fosco, Stephanie Lanza, and SAEED ABDULLAH
DIGITAL HEALTH
DOI: [10.1177/20552076231204418](https://doi.org/10.1177/20552076231204418)

J18 **Busting the one-voice-fits-all myth: Effects of similarity and customization of voice-assistant personality**
Eugene Snyder, Sanjana Mendu, S. Shyam Sundar, and SAEED ABDULLAH
International Journal of Human-Computer Studies
DOI: [10.1016/j.ijhcs.2023.103126](https://doi.org/10.1016/j.ijhcs.2023.103126)

J17 **Technology Based Supportive Care for Metastatic Breast Cancer Patients**
Kathryn Schmitz, Beth Kanski, Brett Gordon, Maxime Caru, Monali Vasakar, Cristina Truica, Ming Wang, Shawna Doerksen, Abby Lorenzo, Renate Winkels, Ling Qiu, and SAEED ABDULLAH
Supportive Care in Cancer
DOI: [10.1007/s00520-023-07884-3](https://doi.org/10.1007/s00520-023-07884-3)

J16 **Preliminary Evaluation of a Conversational Agent to Support Self-management of Individuals Living with PTSD: An Interview Study with Clinical Experts**
Hee Jeong Han, Sanjana Mendu, Beth K Jaworski, Jason E Owen, and SAEED ABDULLAH
JMIR Formative Research
DOI: [10.2196/45894](https://doi.org/10.2196/45894)

- J15 **Women with metastatic breast cancer don't just follow step-count trends, they exceed them: An exploratory study**
Maxime Caru, SAEED ABDULLAH, Ling Qiu, Bethany Kanski, Brett Gordon, Cristina Truica, Monali Vasakar, Shawna Doerksen, and Kathryn Schmitz
Breast Cancer Research and Treatment
DOI: [10.1007/s10549-023-06980-6](https://doi.org/10.1007/s10549-023-06980-6)
- J14 **Addressing metastatic individuals everyday: Rationale and design of the nurse AMIE for Amazon Echo Show trial among metastatic breast cancer patients**
Brett Gordon, Ling Qiu, Shawna Doerksen, Bethany Kanski, Abigail Lorenzo, Cristina Truica, Monali Vasekar, Ming Wang, Renate Winkels, SAEED ABDULLAH, and Kathryn Schmitz
Contemporary Clinical Trials Communications
DOI: [10.1016/j.conctc.2023.101058](https://doi.org/10.1016/j.conctc.2023.101058)
- 2022 J13 **Financial technologies (FinTech) for mental health: The potential of objective financial data to better understand the relationships between financial behavior and mental health**
Johnna Blair, Jeff Brozena, Mark Matthews, Thomas Richardson, and SAEED ABDULLAH
Frontiers in Psychiatry
DOI: [10.3389/fpsyt.2022.810057](https://doi.org/10.3389/fpsyt.2022.810057)
- J12 **Alexa as an Active Listener: How Backchanneling Can Elicit Self-Disclosure and Promote User Experience**
Eugene Cho, Nasim Motalebi, S. Shyam Sundar, and SAEED ABDULLAH
Proceedings of the ACM on Human-Computer Interaction (PACM HCI)
DOI: [10.1145/3555164](https://doi.org/10.1145/3555164)
- J11 **Overcoming Clinician Technophobia: What We Learned from Our Mass Exposure to Telehealth During the COVID-19 Pandemic**
Andrew Sherrill, Christopher Wiese, SAEED ABDULLAH, and Rosa Arriaga
Journal of Technology in Behavioral Science
DOI: [10.1007/s41347-022-00273-3](https://doi.org/10.1007/s41347-022-00273-3)
- J10 **Understanding Design Approaches and Evaluation Methods in mHealth Apps Targeting Substance Use: Protocol for a Systematic Review**
Sahiti Kunchay, Ashley N Linden-Carmichael, Stephanie T Lanza, and SAEED ABDULLAH
JMIR Research Protocols
DOI: [10.2196/35749](https://doi.org/10.2196/35749)
- 2021 J09 **A Query Conundrum: The Mental Challenges of Using a Cognitive Assistant**
Torsten Maier, SAEED ABDULLAH, Christopher McComb, and Jessica Menold
SN Computer Science

DOI: [10.1007/s42979-021-00621-9](https://doi.org/10.1007/s42979-021-00621-9)

- 2020 J08 **“It Didn’t Sound Good with My Cochlear Implants”: Understanding the Challenges of Using Smart Assistants for Deaf and Hard of Hearing Users**
Johnna Blair and SAEED ABDULLAH
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)
DOI: [10.1145/3432194](https://doi.org/10.1145/3432194)
- 2018 J07 **Sensing Technologies for Monitoring Serious Mental Illnesses**
SAEED ABDULLAH and Tanzeem Choudhury
IEEE Multimedia — Special Issue on New Signals in Multimedia
DOI: [10.1109/MMUL.2018.011921236](https://doi.org/10.1109/MMUL.2018.011921236)
- J06 **Personalized stress monitoring: a smartphone-enabled system for quantification of salivary cortisol**
Elizabeth Rey, Aadhar Jain, SAEED ABDULLAH, Tanzeem Choudhury, and David Erickson
Personal and Ubiquitous Computing
DOI: [10.1007/s00779-018-1164-z](https://doi.org/10.1007/s00779-018-1164-z)
- 2017 J05 **Semi-Automated Tracking: A Balanced Approach for Self-Monitoring Applications**
Eun Kyoung Choe, SAEED ABDULLAH, Mashfiqui Rabbi, Edison Thomaz, Daniel Epstein, Matthew Kay, Felicia Cordeiro, Gregory Abowd, Tanzeem Choudhury, James Fogarty, Bongshin Lee, Mark Matthews, and Julie Kientz
IEEE Pervasive Computing
DOI: [10.1109/MPRV.2017.18](https://doi.org/10.1109/MPRV.2017.18)
- 2016 J04 **Development and Evaluation of a Smartphone-Based Measure of Social Rhythms for Bipolar Disorder**
Mark Matthews, SAEED ABDULLAH, Geri Gay, and Tanzeem Choudhury
Assessment (ASM)
DOI: [10.1177/1073191116656794](https://doi.org/10.1177/1073191116656794)
- J03 **Automatic detection of social rhythms in bipolar disorder**
SAEED ABDULLAH, Mark Matthews, Ellen Frank, Gavin Doherty, Geri Gay, and Tanzeem Choudhury
Journal of the American Medical Informatics Association (JAMIA)
DOI: [10.1093/jamia/ocv200](https://doi.org/10.1093/jamia/ocv200)
- 2015 J02 **Mobile Behavioral Sensing for Outpatients and Inpatients With Schizophrenia**
Dror Ben-Zeev, Rui Wang, SAEED ABDULLAH, Rachel Brian, Emily Scherer,

Lisa Mistler, Marta Hauser, John Kane, Andrew Campbell, and Tanzeem Choudhury
Psychiatric Services
 DOI: [10.1176/appi.ps.201500130](https://doi.org/10.1176/appi.ps.201500130)

- 2014 J01 **Tracking Mental Well-Being: Balancing Rich Sensing and Patient Needs**
 Mark Matthews, SAEED ABDULLAH, Geri Gay, and Tanzeem Choudhury
IEEE Computer
 DOI: [10.1109/MC.2014.107](https://doi.org/10.1109/MC.2014.107)

CONFERENCE PAPERS

- 2025 C28 **Thousand voices of trauma: A large-scale synthetic dataset for modeling prolonged exposure therapy conversations**
 Suhas BN, Andrew Sherrill, Rosa Arriaga, Christopher Wiese, and SAEED ABDULLAH *NeurIPS Datasets and Benchmarks Track (spotlight)*
 Acceptance rate: 24.91% • DOI: [10.48550/arXiv.2504.13955](https://doi.org/10.48550/arXiv.2504.13955)
- C27 **The Pursuit of Empathy: Evaluating Small Language Models for PTSD Dialogue Support**
 Suhas BN, Yash Mahajan, Dominik Mattioli, Andrew Sherrill, Rosa Arriaga, Christopher Wiese, and SAEED ABDULLAH *EMNLP (Main Conference)*
 Acceptance rate: 22.16% • DOI: [10.18653/v1/2025.emnlp-main.1573](https://doi.org/10.18653/v1/2025.emnlp-main.1573)
- C26 **How Real Are Synthetic Therapy Conversations? Evaluating Fidelity in Prolonged Exposure Dialogues**
 Suhas BN, Dominik Mattioli, Andrew Sherrill, Rosa Arriaga, Christopher Wiese, and SAEED ABDULLAH *EMNLP (Findings)*
 Acceptance rate: 17.35% • DOI: [10.18653/v1/2025.findings-emnlp.1144](https://doi.org/10.18653/v1/2025.findings-emnlp.1144)
- 2024 C25 **Speaking of Health: Leveraging Large Language Models to assess Exercise Motivation and Behavior of Rehabilitation Patients**
 Suhas BN, Amanda Rebar, and SAEED ABDULLAH
INTERSPEECH
 DOI: [10.21437/Interspeech.2024-121](https://doi.org/10.21437/Interspeech.2024-121)
- C24 **Supportive Fintech for Individuals with Bipolar Disorder: Financial Data Sharing Preferences to Support Longitudinal Care Management**
 Jeff Brozena[†], Johnna Blair[†], Thomas Richardson, Mark Matthews, Dahlia Mukherjee, Erika Saunders, and SAEED ABDULLAH
CHI Conference on Human Factors in Computing Systems
 Acceptance rate: 26.3% • DOI: [10.1145/3613904.3642645](https://doi.org/10.1145/3613904.3642645)
[†]: equal contribution

- 2023 C23 **Differential Privacy enabled Dementia Classification: An Exploration of the Privacy-Accuracy Trade-off in Speech Signal Data**
 Suhas BN, Sarah Rajtmajer, and SAEED ABDULLAH
INTERSPEECH
 DOI: [10.21437/Interspeech.2023-575](https://doi.org/10.21437/Interspeech.2023-575)
- C22 **Knowing How Long a Storm Might Last Makes it Easier to Weather: Exploring Needs and Attitudes Toward a Data-driven and Preemptive Intervention System for Bipolar Disorder**
 Johnna Blair, Dahlia Mukherjee, Erika Saunders, and SAEED ABDULLAH
CHI Conference on Human Factors in Computing Systems
 Acceptance rate: 28.39% • DOI: [10.1145/3544548.3581563](https://doi.org/10.1145/3544548.3581563)
- C21 **Motivation to Use Fitness Application for Improving Physical Activity Among Hispanic Users: The Pivotal Role of Interactivity and Relatedness**
 Maria Molina, Emily Zhan, Devanshi Agnihotri, SAEED ABDULLAH, and Pallav Deka
CHI Conference on Human Factors in Computing Systems
 Acceptance rate: 28.39% • DOI: [10.1145/3544548.3581200](https://doi.org/10.1145/3544548.3581200)
- 2022 C20 **Privacy Sensitive Speech Analysis Using Federated Learning To Assess Depression**
 Suhas BN and SAEED ABDULLAH
The IEEE International Conference on Acoustics, Speech, & Signal Processing (ICASSP)
 Acceptance rate: 45% • DOI: [10.1109/ICASSP43922.2022.9746827](https://doi.org/10.1109/ICASSP43922.2022.9746827)
- 2021 C19 **Assessing effectiveness and interpretability of light behaviors in smart speakers**
 Sahiti Kunchay and SAEED ABDULLAH
Conversational User Interfaces
 Acceptance rate: 29% • DOI: [10.1145/3469595.3469610](https://doi.org/10.1145/3469595.3469610)
- 2020 C18 **Alexa as Coach: Leveraging Smart Speakers to Build Social Agents that Reduce Public Speaking Anxiety**
 Jinping Wang, Hyun Yang, Ruosi Shao, SAEED ABDULLAH, and S. Shyam Sundar
CHI Conference on Human Factors in Computing Systems
 Acceptance rate: 24.31% • DOI: [10.1145/3313831.3376561](https://doi.org/10.1145/3313831.3376561)
- ★ C17 **Will Deleting History Make Alexa More Trustworthy? Effects of Privacy and Content Customization on User Experience of Smart Speakers**
 Eugene Cho, S. Shyam Sundar, SAEED ABDULLAH, and Nasim Motalebi
CHI Conference on Human Factors in Computing Systems
 Acceptance rate: 24.31% • DOI: [10.1145/3313831.3376551](https://doi.org/10.1145/3313831.3376551)

HONORABLE MENTION (TOP 5% OF SUBMISSIONS)

- 2018 C16 **AlertnessScanner: What Do Your Pupils Tell About Your Alertness**
Vincent Tseng, SAEED ABDULLAH, Jean Costa, and Tanzeem Choudhury
Conference on human-computer interaction with mobile devices and services (MobileHCI)
Acceptance rate: 24.5% • DOI: [10.1145/3229434.3229456](https://doi.org/10.1145/3229434.3229456)
- C15 **Understanding Challenges in Prehabilitation for Patients with Multiple Chronic Conditions**
Haining Zhu, Zachary Moffa, Xiyang Wang, SAEED ABDULLAH, Juxihong Julaiti, and John Carroll
Pervasive Health
Acceptance rate: 24% • DOI: [10.1145/3240925.3240959](https://doi.org/10.1145/3240925.3240959)
- 2016 C14 **Cognitive Rhythms: Unobtrusive and Continuous Sensing of Alertness Using a Mobile Phone**
SAEED ABDULLAH, Elizabeth Murnane, Mark Matthews, Matthew Kay, Julie Kientz, Geri Gay, and Tanzeem Choudhury
Conference on Pervasive and Ubiquitous Computing (UbiComp)
Acceptance rate: 23% • DOI: [10.1145/2971648.2971712](https://doi.org/10.1145/2971648.2971712)
- C13 **Shining (Blue) Light on Creative Ability**
SAEED ABDULLAH, Mary Czerwinski, Gloria Mark, and Paul Johns
Conference on Pervasive and Ubiquitous Computing (UbiComp)
Acceptance rate: 23% • DOI: [10.1145/2971648.2971751](https://doi.org/10.1145/2971648.2971751)
- C12 **CrossCheck: Toward passive sensing and detection of mental health changes in people with schizophrenia**
Rui Wang, Min Aung, SAEED ABDULLAH, Dror Ben-Zeev, Rachel Brian, Andrew Campbell, Tanzeem Choudhury, Marta Hauser, John Kane, Michael Merrill, Emily Scherer, and Vincent Tseng
Conference on Pervasive and Ubiquitous Computing (UbiComp)
Acceptance rate: 23% • DOI: [10.1145/2971648.2971740](https://doi.org/10.1145/2971648.2971740)
- ★ C11 **Mobile manifestations of alertness: Connecting biological rhythms with patterns of smartphone app use**
Elizabeth Murnane, SAEED ABDULLAH, Mark Matthews, Matthew Kay, Julie Kientz, Geri Gay, Tanzeem Choudhury, Dan Cosley
Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI)
Acceptance rate: 23.5% • DOI: [10.1145/2935334.2935383](https://doi.org/10.1145/2935334.2935383)
BEST PAPER AWARD (TOP 2 PAPERS)

- C10 **Detecting and Capitalizing on Physiological Dimensions of Psychiatric Illness**
Mark Matthews, SAEED ABDULLAH, Geri Gay, and Tanzeem Choudhury
Conference on Physiological Computing Systems (PhyCS)
DOI: [10.5220/0005952600980104](https://doi.org/10.5220/0005952600980104)
- 2015 C09 **Social (Media) Jet Lag: How Usage of Social Technology Can Modulate and Reflect Circadian Rhythms**
Elizabeth Murnane, SAEED ABDULLAH, Mark Matthews, Tanzeem Choudhury, and Geri Gay
Conference on Pervasive and Ubiquitous Computing (UbiComp)
Acceptance rate: 23% • DOI: [10.1145/2750858.2807522](https://doi.org/10.1145/2750858.2807522)
- C08 **In Situ Design for Mental Illness: Considering the Pathology of Bipolar Disorder in mHealth Design**
Mark Matthews, Stephen Volda, SAEED ABDULLAH, Gavin Doherty, Tanzeem Choudhury, Sangha Im, and Geri Gay
Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI)
Acceptance rate: 27% • DOI: [10.1145/2785830.2785866](https://doi.org/10.1145/2785830.2785866)
- C07 **Collective Smile: Measuring Societal Happiness from Geolocated Images**
SAEED ABDULLAH, Elizabeth Murnane, Jean Costa, and Tanzeem Choudhury
Conference on Computer Supported Cooperative Work & Social Computing (CSCW)
Acceptance rate: 28% • DOI: [10.1145/2675133.2675186](https://doi.org/10.1145/2675133.2675186)
- C06 **MoodLight: Exploring Personal and Social Implications of Ambient Display of Biosensor Data**
Jaime Snyder, Mark Matthews, Jacqueline Chien, Pamara Chang, Emily Sun, SAEED ABDULLAH, and Geri Gay
Conference on Computer Supported Cooperative Work & Social Computing (CSCW)
Acceptance rate: 28% • DOI: [10.1145/2675133.2675191](https://doi.org/10.1145/2675133.2675191)
- 2014 C05 **Towards Circadian Computing: “early to bed and early to rise” makes some of us unhealthy and sleep deprived**
SAEED ABDULLAH, Mark Matthews, Elizabeth Murnane, Geri Gay, and Tanzeem Choudhury
Conference on Pervasive and Ubiquitous Computing (UbiComp)
Acceptance rate: 20% • DOI: [10.1145/2632048.2632100](https://doi.org/10.1145/2632048.2632100)
- 2012 C04 **Towards Population Scale Activity Recognition: A Framework for Handling Data Diversity**
SAEED ABDULLAH, Nicholas Lane, and Tanzeem Choudhury
AAAI Conference on Artificial Intelligence

Acceptance rate: 26% • DOI: [10.5555/2900728.2900849](https://doi.org/10.5555/2900728.2900849)

- 2011 C03 **An Epidemic Model for News Spreading on Twitter**
SAEED ABDULLAH and Xindong Wu
Conference on Tools with Artificial Intelligence (ICTAI)
Acceptance rate: 23% • DOI: [10.1109/ICTAI.2011.33](https://doi.org/10.1109/ICTAI.2011.33)
- 2009 C02 **Evolving Multilayer Neural Networks using Permutation free Encoding Technique**
Anupam Das and SAEED ABDULLAH
Conference on Artificial Intelligence (ICAI)
Acceptance rate: 23%
- 2008 C01 **Permutation Free Encoding Technique for Evolving Neural Networks**
Anupam Das, Shohrab Hossain, SAEED ABDULLAH, and Rashed Ul Islam
International Symposium on Neural Networks (ISNN)
Acceptance rate: 23% • DOI: [10.1007/978-3-540-87732-5_29](https://doi.org/10.1007/978-3-540-87732-5_29)

BOOK CHAPTERS

- 2025 B03 **Technology in Mental Health Work: Challenges and Opportunities**
Lauren Moran, Christopher Wiese, Rosa Arriaga, SAEED ABDULLAH, Andrew Sherrill
Human-Technology Partnerships at Work
In press
- 2021 B02 **Ubiquitous computing for person-environment research: Opportunities, considerations, and future directions**
Sumer Vaid, SAEED ABDULLAH, Edison Thomaz, and Gabriella Harari
Measuring and Modeling Persons and Situations
DOI: [10.1016/B978-0-12-819200-9.00020-X](https://doi.org/10.1016/B978-0-12-819200-9.00020-X)
- 2017 B01 **Circadian Computing: Sensing, Modeling, and Maintaining Biological Rhythms**
SAEED ABDULLAH, Elizabeth Murnane, Mark Matthews, and Tanzeem Choudhury
Mobile Health: Sensors, Analytic Methods, and Applications
DOI: [10.1007/978-3-319-51394-2_3](https://doi.org/10.1007/978-3-319-51394-2_3)

WORKSHOP PAPERS & ABSTRACTS

- 2025 W34 **Designing Collaborative Fintech: Identifying Care Partner Goals and Challenges for Supporting the Financial Lives of Those with Bipolar Disorder**
Hee Jeong Han, Johnna Blair, Jeff Brozena, Dahlia Mukherjee, Erika Saunders,

and SAEED ABDULLAH

Future of Money and HCI workshop at CHI Conference on Human Factors in Computing Systems

W34 **The Currency of Mood: The Use of Open Banking APIs to Better Understand Spending and Bipolar Disorder**

Jeff Brozena, Johnna Blair, Hee Jeong Han, Manar Hesino, Dahlia Mukherjee, Erika Saunders, and SAEED ABDULLAH

Future of Money and HCI workshop at CHI Conference on Human Factors in Computing Systems

W33 **Technoskepticism or Justified Caution? The Future of Human-Centered AI in Mental Health Care**

Nathaniel Swinger, Lauren Moran, SAEED ABDULLAH, Christopher Wiese, Uichin Lee, Andrew Sherrill, and Rosa Arriaga

Panel at CHI Conference on Human Factors in Computing Systems

DOI: [10.1145/3706599.3716286](https://doi.org/10.1145/3706599.3716286)

W32 **Future of Money and HCI**

Johnna Blair, Jeff Brozena, John Vines, Jofish Kaye, Mark Matthews, and SAEED ABDULLAH

Workshop at CHI Conference on Human Factors in Computing Systems

Acceptance rate: 25% • DOI: [10.1145/3706599.3706711](https://doi.org/10.1145/3706599.3706711)

W31 **Voice Assistants to Deliver Cognitive Stimulation Therapy for Persons Living with Dementia**

Ling Qiu, Ita Saragih, Donna Fick, S. Shyam Sundar, and SAEED ABDULLAH

Late-Breaking paper at the CHI Conference on Human Factors in Computing Systems

Acceptance rate: 33.9% • DOI: [10.1145/3706599.3719707](https://doi.org/10.1145/3706599.3719707)

2024 W30 **Using a Human-Centered AI Approach for Managing Finances in Dementia**

Johnna Blair, Ling Qiu, and SAEED ABDULLAH

AAAI Fall Symposium Series 2024 — AI for Aging in Place

W29 **Aging in Place with Dementia: Can Voice Assistants Help Provide Cognitive Stimulation?**

SAEED ABDULLAH, S. Shyam Sundar, Donna Fick, and Ling Qiu

AAAI Fall Symposium Series 2024 — AI for Aging in Place

W28 **Teaming with AI to learn and maintain clinical skills in prolonged exposure: Development of TEAMMAIT (Trustworthy, Explainable, and Adaptive Monitoring Machine for AI Teams)**

Andrew Sherrill, SAEED ABDULLAH, Rosa Arriaga, and Christopher Wiese

W27 Using Objective Financial Data in Mental Health Interventions – Supporting Long-Term Stability and Financial Collaboration for Individuals with Bipolar Disorder

Johnna Blair, Jeff Brozena, Thomas Richardson, Mark Matthews, Dahlia Mukherjee, Erika Saunders, and SAEED ABDULLAH

Association for Behavioral and Cognitive Therapies (ABCT) Annual Convention

W26 Designing financial technologies to support individuals with serious mental illnesses

Johnna Blair, Jeff Brozena, Thomas Richardson, Mark Matthews, Dahlia Mukherjee, Erika Saunders, and SAEED ABDULLAH

Association for Behavioral and Cognitive Therapies (ABCT) Annual Convention

W25 Managing Finances for Persons Living with Dementia: Current Practices and Challenges for Care Partners

Ling Qiu[†], Johnna Blair[†], and SAEED ABDULLAH

Late-Breaking paper at the CHI Conference on Human Factors in Computing Systems

Acceptance rate: 33.88% • DOI: [10.1145/3613905.3650809](https://doi.org/10.1145/3613905.3650809)

†: equal contribution

W24 GAIMPLAY: A Theory-Informed Prompt Engineering Architecture for Personalized Motivational Message Generation to Promote Physical Activity

Sanjana Mendu, Nolan Pye, David Conroy, and SAEED ABDULLAH

Designing (with) AI for Wellbeing workshop at the CHI Conference on Human Factors in Computing Systems

W23 “Learning to Live with Pain”: Designing a Voice Assistant to Deliver Mindfulness-Based Stress Reduction Practices for Individuals Living with Chronic Pain

Sebrina Doyle Fosco, Sanjana Mendu, Stephanie Lanza, and SAEED ABDULLAH

Annual meeting of the Society for Prevention Research

2023 W22 Fintech for Psychological and Financial Resilience: Determinants of Financial Data Sharing Behavior for Individuals with Bipolar Disorder

Jeff Brozena, Johnna Blair, Thomas Richardson, Mark Matthews, Dahlia Mukherjee, Erika Saunders, and SAEED ABDULLAH

Designing for and Reflecting upon Resilience in Health and Wellbeing workshop at Designing Interactive Systems Conference (DIS)

W21 Designing to Support Longitudinal Financial Management and Resilience for Persons Living with Dementia and their Care Partners: Identifying User Needs

and Challenges

Ling Qiu, SAEED ABDULLAH

Designing for and Reflecting upon Resilience in Health and Wellbeing workshop at Designing Interactive Systems Conference (DIS)

- W20 **Designing for and Reflecting upon Resilience in Health and Wellbeing**
Xinning Gui, Yuhan Luo, Xianghua Sharon Ding, SAEED ABDULLAH, Emma Dixon, Shaowen Bardzell
Designing Interactive Systems Conference (DIS)
DOI: [10.1145/3563703.3591456](https://doi.org/10.1145/3563703.3591456)
- W19 **Supportive Data-Driven Financial Technologies for Individuals with Serious Mental Illnesses**
Johnna Blair, Jeff Brozena, Mark Matthews, Thomas Richardson, Dahlia Mukherjee, Erika Saunders, and SAEED ABDULLAH
WISH Symposium at CHI Conference on Human Factors in Computing Systems
In press
- W18 **Impulsive Spending and Finance Management in Bipolar Disorder: Technology as both a Problem and Potential Solution**
Thomas Richardson, Johnna Blair, Jeff Brozena, Mark Matthews, Dahlia Mukherjee, Erika Saunders, and SAEED ABDULLAH
Annual Conference of the International Society for Bipolar Disorders
- W17 **Automatic Classification of Dementia Using Text and Speech Data**
Hee Jeong Han, Suhas B. N., Ling Qiu, and SAEED ABDULLAH
Multimodal AI in Healthcare. Studies in Computational Intelligence
DOI: [10.1007/978-3-031-14771-5_29](https://doi.org/10.1007/978-3-031-14771-5_29)
- 2022 W16 **Comparing Two Delivery Modes Of A Supportive Care Platform For Metastatic Breast Cancer Patients**
Brett Gordon, SAEED ABDULLAH, Shawna Doerksen, Abigail Lorenzo, Bethany Kanski, Ling Qiu, Kathryn Schmitz
Medicine & Science in Sports & Exercise
DOI: [10.1249/01.mss.0000879068.43957.d0](https://doi.org/10.1249/01.mss.0000879068.43957.d0)
- 2021 W15 **PTSDialogue: Designing a Conversational Agent to Support Individuals with Post-Traumatic Stress Disorder**
Hee Jeong Han, Sanjana Mendu, Beth K Jaworski, Jason E Owen, and SAEED ABDULLAH
Mental health: Sensing & Intervention (MHSI) workshop at UbiComp
DOI: [10.1145/3460418.3479332](https://doi.org/10.1145/3460418.3479332)

- W14 **Voice Assistants for Speech Therapy**
 Ling Qiu and SAEED ABDULLAH
Mental health: Sensing & Intervention (MHSI) workshop at UbiComp
 DOI: [10.1145/3460418.3479336](https://doi.org/10.1145/3460418.3479336)
- W13 **Nurse AMIE: Using Smart Speakers to Provide Supportive Care Intervention for Women with Metastatic Breast Cancer**
 Ling Qiu, Bethany Kanski, Shawna Doerksen, Renate Winkels, Kathryn H.Schmitz, and SAEED ABDULLAH
Late-Breaking paper at the CHI Conference on Human Factors in Computing Systems
 Acceptance rate: 39% • DOI: [10.1145/3411763.3451827](https://doi.org/10.1145/3411763.3451827)
- W12 **Using Tweets to Assess Mental Well-being of Essential Workers During the COVID-19 Pandemic**
 Johnna Blair, Chi-Yang Hsu, Ling Qiu, Shih-Hong Huang, Ting-Hao (Kenneth) Huang, and SAEED ABDULLAH
Late-Breaking paper at the CHI Conference on Human Factors in Computing Systems
 Acceptance rate: 39% • DOI: [10.1145/3411763.3451612](https://doi.org/10.1145/3411763.3451612)
- W11 **Context-Specific Usability Measures for Voice Assistants**
 Sanjana Mendu, S. Shyam Sundar, and SAEED ABDULLAH
"Let's Talk About CUIs: Putting Conversational User Interface Design Into Practice" workshop at the CHI Conference on Human Factors in Computing Systems
- 2020 W10 **Opportunities for Collaborative Clinical Work: Predicting Relapse Onset in Bipolar Disorder from Online Behavior**
 Johnna Blair, Erika Saunders, Dahlia Mukherjee, and SAEED ABDULLAH
Pervasive Health
 DOI: [10.1145/3421937.3421947](https://doi.org/10.1145/3421937.3421947)
- W09 **WatchOver: using Apple watches to assess and predict substance co-use in young adults**
 Sahiti Kunchay and SAEED ABDULLAH
Mental health: Sensing & Intervention (MHSI) workshop at UbiComp
 DOI: [10.1145/3410530.3414373](https://doi.org/10.1145/3410530.3414373)
- 2019 W08 **Investigating Users' Perceptions of Light Behaviors in Smart-Speakers**
 Sahiti Kunchay, Shan Wang, and SAEED ABDULLAH
Conference on Computer Supported Cooperative Work & Social Computing (CSCW)
 DOI: [10.1145/3311957.3359479](https://doi.org/10.1145/3311957.3359479)

- W07 **Can Alexa be your Therapist? How Back-Channeling Transforms Smart-Speakers to be Active Listeners**
 Nasim Motalebi, Eugene Cho, S Shyam Sundar, and SAEED ABDULLAH
Conference on Computer Supported Cooperative Work & Social Computing (CSCW)
 DOI: [10.1145/3311957.3359502](https://doi.org/10.1145/3311957.3359502)
- W06 **Understanding the Needs and Challenges of Using Conversational Agents for Deaf Older Adults**
 Johnna Blair and SAEED ABDULLAH
Conference on Computer Supported Cooperative Work & Social Computing (CSCW)
 DOI: [10.1145/3311957.3359487](https://doi.org/10.1145/3311957.3359487)
- 2018 W05 **Conversational agents to provide couple therapy for patients with PTSD**
 Nasim Motalebi and SAEED ABDULLAH
WellBeCoach: Smart Coaching Solutions for Health & Wellbeing workshop at Pervasive Health
 DOI: [10.1145/3240925.3240933](https://doi.org/10.1145/3240925.3240933)
- W04 **Supporting Constructive Mental Health Discourse in Social Media**
 Johnna Blair and SAEED ABDULLAH
Design4Diversity workshop at Pervasive Health
 DOI: [10.1145/3240925.3240930](https://doi.org/10.1145/3240925.3240930)
- 2016 W03 **Assessing mental health issues on college campuses: preliminary findings from a pilot study**
 Vincent Tseng, SAEED ABDULLAH, Michael Merrill, Min Aung, Franziska Wittleder, and Tanzeem Choudhury
Mental health: Sensing & Intervention (MHSI) workshop at UbiComp
 DOI: [10.1145/2968219.2968308](https://doi.org/10.1145/2968219.2968308)
- 2015 W02 **Towards circadian computing: a sensing & intervention framework for body clock friendly technology**
 SAEED ABDULLAH
Doctoral Colloquium at UbiComp
 DOI: [10.1145/2800835.2801657](https://doi.org/10.1145/2800835.2801657)
- W01 **Biological rhythms and technology**
 Mark Matthews, Erin Carroll, SAEED ABDULLAH, Jaime Snyder, Matthew Kay, Tanzeem Choudhury, Geri Gay, Julie Kientz
CHI Conference on Human Factors in Computing Systems
 DOI: [10.1145/2559206.2559230](https://doi.org/10.1145/2559206.2559230)

POSTERS & DEMOS

- 2025 ★ P16 **Generative AI for Efficient and Empathetic Conversational Models in Mental Health**
 Suhas BN, Dominik Mattioli, Rosa Arriaga, Christopher Wiese, Andrew Sherrill, and SAEED ABDULLAH
Penn State AI Week 2025
 3RD BEST POSTER AWARD
- P15 **Generative AI for Clinician-Supported Exposure Therapy: Toward Personalized Script and Image Generation**
 Sam Moradzadeh and SAEED ABDULLAH
Penn State AI Week 2025
- ★ P14 **Generative AI for Efficient and Empathetic Conversational Models in Mental Health**
 Suhas BN, Dominik Mattioli, Rosa Arriaga, Christopher Wiese, Andrew Sherrill, and SAEED ABDULLAH
Mid-Atlantic Student Colloquium on Speech, Language and Learning (MASC-SLL 2025)
 BEST POSTER AWARD
- 2023 P13 **Development of a Voice Interface to Support Mindfulness Practice for Individuals with Chronic Pain**
 Sanjana Mendu, Sebrina Doyle Fosco, Stephanie Lanza, and SAEED ABDULLAH
Consortium on Substance Use and Addiction (CSUA) Annual Conference
- 2022 P12 **Dementia Diagnosis using Text and Speech Data**
 Hee Jeong Han, Suhas BN, Ling Qiu, and SAEED ABDULLAH
ICDS Fall 2022 Symposium: Data Science, AI, and a Sustainable, Resilient, and Equitable Future
- P11 **ACOUSTICS : Automatic classification of subjects with dementia and healthy controls using text transcriptions and speech data**
 Hee Jeong Han, Suhas BN, Ling Qiu, and SAEED ABDULLAH
Health Intelligence (W3PHIAI-22) Workshop at the AAAI Conference on Artificial Intelligence
- 2015 P10 **Automatic detection of social rhythms in bipolar disorder via smartphone**
 Ellen Frank, SAEED ABDULLAH, Mark Matthews, and Tanzeem Choudhury
American College of Neuropsychopharmacology (ACNP)
- P09 **Circadian Computing: Towards bodyclock friendly technology**
 SAEED ABDULLAH

Human Computer Interaction Consortium Workshop (HCIC)

- P08 **SAINT: a scalable sensing and inference toolkit**
Mashfiqui Rabbi, Thiago Caetano, Jean Costa, SAEED ABDULLAH, Mi Zhang,
and Tanzeem Choudhury
Workshop on Mobile Computing Systems and Applications (ACM HotMobile)
- 2014 P07 **Towards circadian computing: “early to bed and early to rise” makes some of us unhealthy and sleep deprived**
SAEED ABDULLAH, Mark Matthews, Elizabeth Murnane, Geri Gay, and Tanzeem Choudhury
Intel Science and Technology Center for Pervasive Computing (ISTC-PC)
- P06 **Circadian Computing: Towards bodyclock friendly technology**
SAEED ABDULLAH
Human Computer Interaction Consortium Workshop (HCIC)
- 2013 P05 **Developing a smartphone app to monitor mood, social rhythms, sleep and social activity: technology to support effective management of bipolar disorder**
Ellen Frank, Mark Matthews, Tanzeem Choudhury, Steve Volda, and SAEED ABDULLAH
American College of Neuropsychopharmacology (ACNP)
- 2013 P04 **Light, color, affect, and stress**
Jaime Snyder, Mark Matthews, SAEED ABDULLAH, Yohan Ko, and Geri Gay
Society for Social Studies of Science (4S)
- P03 **MoodRhythm: Tracking and supporting daily rhythms**
Stephen Volda, Mark Matthews, SAEED ABDULLAH, Mengxi Chrissie Xi, Matthew Green, Won Jun Jang, Donald Hu, John Weinrich, Prashama Patil, Mashfiqui Rabbi, Tauhidur Rahman, Geri Gay, Ellen Frank, and Tanzeem Choudhury
Interactive demo at UbiComp
- P02 **Clockwise: inferring chronotype and daily patterns from smartphone use**
SAEED ABDULLAH
Doctoral Colloquium at UbiComp
- 2012 P01 **Towards population scale activity recognition: a framework for handling data diversity**
SAEED ABDULLAH, Nicholas D. Lane, and Tanzeem Choudhury
Intel Science and Technology Center for Pervasive Computing (ISTC-PC)

TEACHING

- 2025 **Human Centered Artificial Intelligence (HCAI)**
College of Information Sciences and Technology, Penn State
Developed syllabus, designed instruction material
- 2021 – **HCDD 340: Human-Centered Design for Mobile Computing**
College of Information Sciences and Technology, Penn State
Co-designed curriculum, developed syllabus, designed instruction material
- 2018 – 2024 **IST 525: Computer Supported Cooperative Work (CSCW)**
College of Information Sciences and Technology, Penn State
Developed syllabus, designed instruction material
- 2019 – 2021 **IST 331: Foundations of Human-Centered Design**
College of Information Sciences and Technology, Penn State
Developed syllabus, designed instruction material
- 2017 – 2018 **IST 597: Computational Health: Sensing and Intervention Design**
College of Information Sciences and Technology, Penn State
Developed syllabus, designed instruction material
- 2014 – 2016 **Info 4120/6120: Ubiquitous Computing**
Department of Information Science, Cornell University. Instructor: Tanzeem Choudhury
Delivered a guest lecture on Circadian Computing and led a lab session on mobile programming
- Spring, 2012 **CS/Info 2300: Intermediate design and programming for the web**
Department of Information Science, Cornell University. Instructor: Carl Lagoze.
Led lab teaching sections and covered some lectures
- Fall, 2012 **CS 4300: Information retrieval**
Department of Information Science, Cornell University. Instructor: Paul Ginsberg.
Tutored students and graded assignments
- Spring, 2011 **CS 224: Algorithm design & analysis**
University of Vermont. Instructor: Byung Lee
Covered some lectures, tutored students, and marked assignments
- Fall, 2010 **CS 32: Puzzles, games and algorithms**
University of Vermont. Instructor: Robert Snapp
Led lab sessions and marked assignments

CS 204: Database system

University of Vermont. Instructor: Byung Lee

Covered some lectures, tutored students, and marked assignments

Spring, 2010

CS 222: Computer architecture

University of Vermont. Instructor: Alan Ling

Tutored students and marked assignments

CS 195: Computer science for geo-spatial technologies

Instructor: Alison Pechenick

Developed and marked assignments

Fall, 2009

CS 201: Operating systems

University of Vermont. Instructor: Alan Ling

Covered some lectures, tutored students, and marked assignments

CS 243: Theory of computation

University of Vermont. Instructor: Alan Ling

Tutored students and marked assignments

RESEARCH SUPPORT

2025 – 2030

DeepStroke+: An Advanced Mobile AI Diagnostic Tool for Fast and Precise Detection of Acute Strokes in Mobile Stroke Units, Emergency Rooms, and Telestroke Triage

Co-Investigator

National Institute of Neurological Disorders and Stroke, (\$1,557,560)

2024 – 2029

Penn State Roybal Center for Promoting Adherence to Behavior Change and Enhancing Cognitive Function

Co-Investigator

National Institute on Aging, (total: \$5,698,199; at Penn State: \$1,370,559)

2024 – 2026

Voice Assistant to Provide Cognitive Stimulation Therapy at Scale

Principal Investigator

National Institute on Aging, (\$432,198)

2023 – 2027

Collaborative Research: FW-HTF-RL: Understanding the Ethics, Development, Design, and Integration of Artificial Intelligence Teammates in Future Mental Health Work

Principal Investigator at Penn State

National Science Foundation (total: \$1,999,957; at Penn State: \$650,000)

- 2023 – 2025 **Financial Activity Data as an Objective Behavioral Marker in Bipolar Disorder: A Feasibility and Acceptance Study**
Principal Investigator
National Institute of Mental Health (\$437,685)
- 2022 – 2024 **Financial Wellbeing for Individuals Living with Mental Illnesses: Toward a Privacy-preserving and Data-driven Intervention Platform**
Principal Investigator
Social Science Research Institute (SSRI), Penn State (\$75,000)
- 2019 – 2023 **CRII:SCH: Designing for Supportive Accountability: Using Conversational Agents to Sustain Patient Engagement in PTSD**
Principal Investigator
National Science Foundation (\$174, 818)
- 2019 – 2021 **Nurse AMIE (Addressing Metastatic Individuals Everyday)**
Co-Investigator
American Institute for Cancer Research (\$165,000)
- 2018 – 2021 **A Real-Time Mindfulness Intervention to Control Pain: Delivery Through a Conversational Agent**
Principal Investigator
Social Science Research Institute (SSRI), Penn State (\$20, 000)
- 2018 – 2021 **Predicting Relapse Onset in Bipolar Disorder from Online Behavioral Data**
Principal Investigator
Institute for CyberScience (ICS), Penn State (\$37, 500)
- 2019 – 2020 **Watch Over: Using Apple Watches to Assess and Predict Substance Co-use in Young Adults**
Principal Investigator
College of Information Sciences and Technology, Penn State (\$53, 260)
- 2018 – 2019 **Talking to Machines: Virtual Conversational Agents for Effective Treatment Delivery in PTSD**
Principal Investigator
College of Information Sciences and Technology, Penn State (\$56, 310)

AWARDS

- 2025 **Best Poster Award**
Mid-Atlantic Student Colloquium on Speech, Language and Learning (MASC-SLL)

2025)

For the poster “Generative AI for Efficient and Empathetic Conversational Models in Mental Health” (out of 62 posters).

3rd Best Poster Award

Penn State AI Week 2025

For the poster “Generative AI for Efficient and Empathetic Conversational Models in Mental Health” (out of 80+ posters).

2023 **Dean’s Circle of Teaching Excellence**

College of Information Sciences and Technology, Penn State

This award recognizes “contributions to our teaching mission based on a broad definition of teaching which includes classroom teaching, mentoring of students, curriculum development, and other teaching related activities.”

2021 **SIGCHI Development Award**

ACM SIGCHI

This award (\$3050) focused on improving inclusivity and diversity in Ubi-Comp and ISWC conference. I developed the proposal as a scholarship co-chair of the conference.

2020 **Honorable Mention**

CHI Conference on Human Factors in Computing Systems

For the paper: “Will Deleting History Make Alexa More Trustworthy? Effects of Privacy and Content Customization on User Experience of Smart Speakers”

Distinguished Master’s Thesis Award

Graduate School, Penn State

Awarded to Shan Wang (my advisee) for “excellence in master’s level thesis research”. Her thesis title is: “Effects of Prompt Humor Level on Behavioral Intention and Perceived Interruption: An Online Survey Study”.

2019 **Junior Faculty Excellence Award in Research**

College of Information Sciences and Technology, Penn State

This award recognizes “innovative and outstanding research results, contributed to breakthrough and high impact research, or demonstrated the potential to solve challenging research problems”.

2018 **Tronzo Medical Informatics Endowment**

College of Information Sciences and Technology, Penn State

The endowment supports research on issues and technologies related to medical informatics.

2016 – **Special recognition for outstanding reviews**

CHI 2021

CHI 2020

IMWUT (February 2020)

IMWUT (November 2019)

IMWUT (May 2017 round)

UbiComp 2016

2016 **Best Paper Award**

Conference on human-computer interaction with mobile devices and services (MobileHCI)

For the paper: Mobile manifestations of alertness: Connecting biological rhythms with patterns of smartphone app use.

Agile Research Grant

Robert Wood Johnson Foundation

Our project focusing on circadian rhythms and cognitive performance was selected as one of the five recipients in the Agile Research Project competition.

2013 **\$100,000 Heritage Open mHealth Challenge winner**

Heritage Provider Network, Open mHealth, and the University of California, Los Angeles

Our MoodRhythm project focusing on bipolar disorder won the Heritage Open mHealth Challenge.

INVITED TALKS

2025 **Following the Money: How Financial Data Can Transform Mental Health Research and Care**

AI for Social Impact Seminar Series at CSRAI Penn State

Following the Money: How Financial Data Can Transform Mental Health Research and Care

Workshop on Financial Technology and Natural Language Processing (FinNLP)

Following the Money: How Financial Data Can Transform Mental Health Research and Care

Dartmouth Center for Technology and Behavioral Health (CTBH) Seminar Series

2024 **Write On: Opportunities and Challenges in using Generative AI for Health Intervention Messaging**

BRiTE Center, University of Washington

Write On: Opportunities and Challenges in using Generative AI for Health Intervention Messaging

Closing keynote at annual conference, UConn Center for mHealth and Social Media

Building Better Digital Health: A Data-Driven and Human-Centered Approach

Millennium Café, Penn State

2023 **Talking to Machines: Designing Conversational Agents to Support Health Intervention**

SPSP Psychology of Media and Technology preconference

2021 **Talking to Machines: Leveraging Voice Interfaces for Intervention Delivery**

Center for Behavioral Health and Technology, University of Virginia

“It Didn’t Sound Good with My Cochlear Implants”: Understanding the Challenges of Using Smart Assistants for Deaf and Hard of Hearing Users

Johnna Blair and SAEED ABDULLAH

National Educational Conference, Canadian Hard of Hearing Association

2020 **Designing for Supportive Accountability: Using Conversational Agents to Sustain Patient Engagement in Post-traumatic stress disorder (PTSD)**

Advancing Health through Science Smart and Connected Health (SCH) Principal Investigator Meeting, National Science Foundation

2019 **Computational Health: Using HCI & Data Science Methods to Improve Health and Wellbeing**

Dean’s Advisory Board, College of Information Sciences and Technology, Penn State

“Alexa, I am in Pain!”: A Real-Time Mindfulness Intervention for Chronic Pain Management Using Conversational Agents

Prevention Research Center (PRC) Day, College of Health and Human Development, Penn State

2018 **Your phone usage reveals a lot about you: Inferring sleep and circadian disruptions from phone use patterns**

QuantDev Brownbag Series, College of Health and Human Development, Penn State

Talking to Machines: Conversational Agents for Mental Health Care

Colloquium at UT Austin iSchool

Talking to Machines: Conversational Agents for Mental Health Care

Grand Rounds Series, Department of Psychology at Penn State

- 2016 **Passive sensing of circadian rhythms for individualized models of cognitive performance**
Health Data Exploration Project Agile Grant Series
- 2015 **Cognitive rhythms: unobtrusive and continuous sensing of alertness using a mobile phone**
Intel Science & Technology Center for Pervasive Computing (ISTC-PC)
- 2014 **Supporting individuals with bipolar disorder to establish stable daily routines**
MIT & Cornell hacking-medicine hackathon

PROFESSIONAL EXPERIENCE

- 2012 – 2017 **Graduate Research Assistant**
Department of Information Science, Cornell University. Supervisor: Tanzeem Choudhury
- Fall, 2015 **Research Intern**
Microsoft Research. Supervisor: Mary Czerwinski
Designed and developed novel system for improving creative ability
- 2011 – 2012 **Graduate Teaching Assistant**
Department of Information Science, Cornell University
- 2009 – 2011 **Graduate Teaching Assistant**
University of Vermont
- 2008 – 2009 **Software Developer**
AfriGIS Ltd
Developing search engine with special focus on geographic relevance
- 2008 **Software Developer**
Google Summer of Code
Enhancing a Just-In-Time (JIT) compiler for Java to provide instruction code selection and emission for array manipulation bytecodes

SERVICE

EDITORIAL RESPONSIBILITIES

- 2022 **Topic Editor**
Frontiers Research Topic: "Factors Influencing User Engagement with Digital Men-

tal Health Interventions”

2021 **Topic Editor**

Frontiers Research Topic: “Quantitative Modeling of Psychopathology Using Passively Collected Data”

2017 – 2020 **Associate Editor**

Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)

2018 **Co-Editor**

Special Issue of JMIR on Computing and Mental Health

CONFERENCE & WORKSHOP ORGANIZER

2025 **Workshop on “the Future of Money and HCI”**

CHI Conference on Human Factors in Computing Systems

Panel on “Technoskepticism or Justified Caution? The Future of Human-Centered AI in Mental Health Care”

CHI Conference on Human Factors in Computing Systems

2016 – **Workshop on “Mental health and well-being: Sensing and intervention”**

Conference on Pervasive and Ubiquitous Computing (UbiComp)

2023 **Designing for and Reflecting upon Resilience in Health and Wellbeing**

Designing Interactive Systems (DIS)

2022 **Award & Recognition Committee member**

Computer Supported Cooperative Work & Social Computing (CSCW)

2021 **Scholarship co-chair**

Conference on Pervasive and Ubiquitous Computing (UbiComp)

2019 **Publicity co-chair**

Conference on Pervasive and Ubiquitous Computing (UbiComp)

2014 **Workshop on “Biological rhythms and technology”**

CHI Conference on Human Factors in Computing Systems

PROGRAM COMMITTEE MEMBER

- 2016 – **Workshop on “Mental health and well-being: Sensing and intervention”**
Conference on Pervasive and Ubiquitous Computing (UbiComp)
- 2017 – 2020 **Pervasive Health**
EAI International Conference on Pervasive Computing Technologies for Healthcare
- 2015 – 2018 **International Conference on Digital Health**
- 2017 **Workshop on “Tools and Algorithms for Mental Health and Wellbeing, Pain, and Distress (MHWPDP)”**
Affective Computing and Intelligent Interaction (ACII)

ACADEMIC SERVICES

- 2025 – 2026 **Departmental Promotion and Tenure Committee member**
College of Information Sciences and Technology, Penn State
- 2025 – 2026 **Department Head Search Committee member**
College of Information Sciences and Technology, Penn State
- 2024 – 2025 **Chair of the Faculty Search Committee (Human-Centered Artificial Intelligence track)**
College of Information Sciences and Technology, Penn State
- 2022 – 2023 **Faculty Search Committee (Human-Centered Artificial Intelligence track)**
College of Information Sciences and Technology, Penn State
- 2023 **HCI Faculty Annual Evaluation Committee member**
College of Information Sciences and Technology, Penn State
- 2021 – 2022 **Chair of the Faculty Search Committee (Accessibility track)**
College of Information Sciences and Technology, Penn State
- 2020 – 2022 **Academic Integrity Committee**
College of Information Sciences and Technology, Penn State
- 2019 – **Advisory Board Member**
Penn State SELF (Student Engagement, Learning, and Flourishing) project

- 2017 – **Governing Board Member**
Human Computer Interaction Consortium (HCIC)
- 2017 – **Faculty advisory committee focusing on IT support for research**
College of Information Sciences and Technology, Penn State
- 2018 – 2020 **Faculty Search Committee (HCD track)**
College of Information Sciences and Technology, Penn State
- Graduate Recruiting Committee**
College of Information Sciences and Technology, Penn State
- 2020 **Application Reviewer**
Nittany AI Challenge, Penn State
- 2017 **Application Reviewer**
Schreyer Honors College, Penn State
- Applicant Interviewer**
Millennium Scholars program, Penn State
- 2015 **Student Volunteer**
Program committee meeting for Conference on Pervasive and Ubiquitous Computing (UbiComp)
- 2013 – 2015 **Student Representative**
Faculty Search Committee, Department of Information Science, Cornell University
- 2011 – 2013 **Colloquium Organizer**
Department of Information Science, Cornell University
- 2011 – 2012 **Computing facilities support**
Department of Information Science, Cornell University